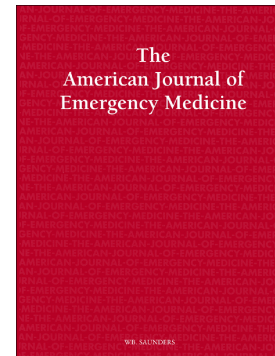


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Ketamine versus Ketamine plus atropine for pediatric sedation: a meta-analysis

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Abstract

Objectives: The application of atropine for pediatric sedation in the emergency department remains controversial. Our objective was to perform a comprehensive review of the literature and assess the clinical indexes in groups with and without atropine use.

Methods: PubMed, EMBASE, and the Cochrane Library were searched for randomized and non-randomized studies that compared ketamine and ketamine plus atropine for pediatric sedation. The risk ratio with 95% confidence interval was calculated using either a fixed- or random-effects model according to the value of I^2 .

Results: One retrospective study and four randomized controlled trials were identified to compare the clinical indexes. For the clinical indexes, the ketamine plus atropine group had better outcomes than the ketamine group in hypersalivation ($P < 0.05$), but indexes of rash and tachycardia were worse. The two methods of sedation were comparable for nausea, vomiting, desaturation, agitation and laryngospasm ($P > 0.05$).

Conclusions: Based on the current evidence, the group receiving atropine had reduced hypersalivation and increased rash and tachycardia; no differences were observed in nausea, vomiting, desaturation, agitation and laryngospasm between the two groups. Given that some of the studies were of low quality, additional high-quality randomized controlled trials should be conducted to further verify these findings.

Introduction

As an important department of the hospital, the emergency department (ED) sees many patients daily, and children account for a large proportion of these patients. Children and adults feel similarly when they face trauma and pain. However, children are less able to adapt to the emergency environment than adults, which can lead to increased anxiety and pain.[1] What is more, when anxiety spreads among children

and parents, the child's condition can worsen.[2] Procedural sedation and analgesia (PSA) is very important and can reduce unnecessary anxiety during the examination. At the same time, the selection of sedative drugs and, when multiple agents are used, drug interactions, can be challenging for emergency doctors.

Ketamine is a drug widely used for sedation and analgesia in emergency departments in many countries. It is a phencyclidine nonbarbiturate derivative, and its effects are achieved by combining sigma opioid receptors and N-methyl-d-aspartate.[3] Compared with other drugs, ketamine has the advantage of being fast acting with easy recovery. It exhibits excellent safety when used by non-anesthesiologists.[4,5] However, it has some side effects, including nausea, vomiting, agitation, transient rash, and hypersalivation.[1,4,6] Atropine is a well-known antimuscarinic drug[7] and is widely used to limit excessive mucosal secretions.[8] However, use of atropine delays the onset of saliva, which is itself a complication.[9] Therefore, whether atropine should be combined with ketamine to calm children is controversial, and many related experiments are being tested to investigate the problem.[9,10] Therefore, we collected articles about atropine use in ketamine sedation and conducted a meta-analysis to provide evidence to help guide the doctor's decision.

Methods –

Search strategy

Two investigators reviewed the literature using the PICO principles, which include four elements: "P" refers to the patient, population or problem; "I" is the intervention; "C" stands for comparison; and "O" is the outcome. The key words ketamine, atropine, sedation and similar words were searched as [MESH] terms. The Boolean operators "AND" and "OR" were used to connect terms and search the literature in PubMed, EMBASE, and the Cochrane Library. The search was not limited to an initial time and language, but the deadline established was January 3,

2018. To avoid the omission of relevant documents, the researchers did not stipulate a patient population and selected the “All Fields” option rather than “Title/Abstract”. In the rest of our work, type of population was limited by selection criteria. Based on the titles and abstracts, the researchers selected potentially eligible studies and read the full text of selected articles to assess eligibility. Disagreements between the two reviewers were decided by a third individual.

Selection criteria

(1) Participants: Patients aged less than 16 years old who were administered pediatric procedural sedation with ketamine in the emergency department were considered to meet the inclusion criteria. Patients over the age of 16 were excluded.

(2) Intervention and comparison: The group of patients who received ketamine with adjunctive atropine was the intervention group, and patients who received ketamine only or ketamine plus water or saline were placed in the control group. Patients who had received another drug for sedation were excluded.

(3) Outcomes: The clinical indexes, including vomiting, nausea, desaturation, hypersalivation, rash, tachycardia, agitation and muscle laryngospasm, were used as the outcomes.

(4) Study design: Randomized controlled trials and retrospective studies that compared ketamine with ketamine plus placebo were considered qualified.

Quality assessment

We used the Cochrane Handbook to evaluate the risk of randomized controlled trial data, and, according to the results, we classified the studies as high risk, low risk and unclear risk. In addition, for the retrospective study, we used the Newcastle-Ottawa Scale to evaluate the article quality, i.e., classify the trials into three levels of quality.

Data extraction

We extracted data on first author, publication date, country, study design, number of patients, mean age, percentage by sex, mean weight (kg), ASA scores and

interventions. When disagreement occurred, the third reviewer made the final decision.

Data analysis and statistical methods

We used Review Manager version 5.3 to analyze the data. The risk ratio (RR) was calculated for the dichotomous outcomes. We used I^2 values to assess heterogeneity among the articles. If $I^2 > 50$, we used the random-effect model. If the opposite, we chose the fixed-effect model.

Results

Search results

A total of 545 articles were identified by our query method, of which 73 were from PubMed, 392 from EMBASE, and 80 from the Cochrane Library. Of these articles, 122 studies were eliminated as duplicates. Investigators selected 365 articles based on the meaning of the title and abstract. Finally, we chose five articles after considering the full text. The whole document screening process is reflected in Fig. 1.

Risk of bias assessment

In these included documents, the methodological quality of two types of experiments was evaluated according to their respective evaluation criteria. Only one the randomized controlled trials (RCTs) was low risk[11]; the rest were high risk. In the studies on RCTs that had low risk of reporting bias, only 3[11-13] reported random sequence generation, and 3[9,11,13] had a low risk for allocation concealment, binding of outcome assessment, and mention of participants and personnel in the text. For other biases, we were able to find clues in the texts. The only retrospective study was considered to be of good quality. Details about its contents are exhibited in Fig. 2a, 2b and Table 1.

Study characteristics

Of the 5 articles included, 4 studies were RCTs, and 1 article was a retrospective

study. These articles described single-center studies, and nearly every article provided the general characteristics of the study population. In four articles, the patient's condition was evaluated by ASA; only one did not refer to an assessment of the patient's condition. Finally, 969 people were included in our study. Among them, 445 patients were sedated with ketamine, and 524 patients received ketamine plus atropine. The characteristics of these documents are presented in Table. 2

Outcomes of meta-analysis

1.1.1 Nausea

Two reports provided data (n=340) on nausea. A fixed-effects model was used, and no significant heterogeneity was found ($I^2=0\%$, $P=0.41$). The incidence of nausea in the ketamine + atropine group was not lower than that in the ketamine group ($RR=0.81$, 95% CI: 0.45-1.46, $P=0.49$). (Fig. 3)

1.1.2 Vomiting

There are five reports offering data (n=954) on vomiting. Of them, 4 studies were randomized controlled trials and 1 study was a retrospective study. A fixed-effects model was used, and heterogeneity was slight ($I^2=11\%$, $P=0.35$). Therefore, additional administration of atropine had no obvious inhibitory effect on vomiting after ketamine sedation ($RR=0.74$, 95% CI: 0.53-1.03, $P=0.07$). (Fig. 4)

1.1.3 Desaturation

Three reports reported the oxygen desaturation. We used a fixed model, and no significant heterogeneity was found ($I^2=0\%$, $P=0.84$). The advantage of ketamine + atropine was not shown ($RR=0.92$, 95% CI: 0.26-3.22, $P=0.90$). (Fig. 5)

1.1.4 Hypersalivation

Three studies with 423 patients reported on the symptom of hypersalivation. A fixed model was used, and no significant heterogeneity was found ($I^2=0\%$,

P=0.63). The rate of hypersalivation that occurred in the ketamine + atropine group was lower than that in the ketamine group (RR=0.37, 95% CI: 0.23-0.62, P=0.0001). (Fig. 6)

1.1.5 Rash

The incidence of rash was provided in two reports. A fixed-effects model was used, and a small amount of heterogeneity was found ($I^2=19\%$, $P=0.27$). The incidence of rash in the experimental group was higher than that in control group (RR=2.44, 95% CI: 1.05-5.71, $P=0.04$). (Fig. 7)

1.1.6 Tachycardia

Two articles with 340 patients reported the outcome of tachycardia. A fixed-effects model was used, and no significant heterogeneity was found ($I^2=0\%$, $P=0.68$). The results showed that the tachycardia in the ketamine + atropine group was significantly higher than that in the ketamine group (RR=2.22, 95% CI: 1.35-3.67, $P=0.002$) (Fig. 8).

1.1.7 Agitation

Data regarding agitation were available in two reports with 223 patients. A fixed-effects model was used, and no significant differences were observed between the two groups ($I^2=0\%$, $P=0.69$, RR=0.83, 95% CI: 0.38-1.79, $P=0.63$) (Fig. 9).

1.1.8 Laryngospasm

Two reports contributed to the analysis of laryngospasm. A fixed-effects model was used, and no significant heterogeneity was found ($I^2=0\%$, $P=0.40$). The result proved that the occurrence of laryngospasm is not associated with the addition of atropine (RR=0.85, 95% CI: 0.36-1.99, $P=0.70$). (Fig. 10)

Discussion

With constant research and practice, procedural sedation of children in the emergency department is safe and effective[14] , besides it has been slightly improved.[15] For injured children, the ability providing safe and effective sedation and analgesic is an essential skill for non-anesthesiologists in the emergency department[16]. From the development of the first sedation guideline to the establishment of multiple organizations and guideline drafts, every step reflects progress toward helping doctors perform better.[17-21] Furthermore, emergency doctors would also have relevant trainings, making them familiar with the characteristics of the selected drug, including dose, duration, etc. Based on these conditions, plus non-invasive monitoring and preoperative evaluation, the success rate of the PSA is again greatly improved[16,22]. For drug safety, a prospective descriptive study including 1215 patients suggested the success rate of sedation up to 98.6%[23], , and the other study proved that adverse effects were common, but the serious adverse effects rarely occurred after 25min from the final medication administration[24]. According to statistics, death events in the ED due to the PSA are more rare.[25] We should aim to prevent the death of patients but also pay attention to other aspects of the PSA. We had known of the adverse reactions on ketamine, which atropine can treat effectively. However, the drug mechanism of atropine is antimuscarinic, and its range of action was large. Therefore, we used a meta-analysis to estimate the safety and effectiveness of atropine.

In our analysis, reliable evidence was supplied through summarized evidence from randomized and other types of trials. As outcome parameters, improvements in hypersalivation were demonstrated with the addition of atropine compared with use of ketamine alone. However, rash and tachycardia showed the disadvantages of the combination with atropine. Next, we will explain the reasons underlying each observation. Let us first examine the unique advantage that atropine demonstrated in this analysis. This result was consistent with 3 studies[9,11,12]. In the study of Kye et al, we knew that 11.4% of patients using atropine still had hypersalivation, and this outcome of the control group was idiosyncratic.[10,12] For hypersalivation, except for those children needing suction and airway repositioning, most patients rarely needed intervention.[12] However, this did not mean that atropine was of little use in sedation; Heinz et al concluded that atropine could reduce the incidence of postprocedural vomiting. Second, we sought to explore the relationship between atropine and states of illness for rash and tachycardia. Rash was a common complication of atropine combined with ketamine, which usually was not serious and would abate without intervention.[9] Unlike rash, the main reason for tachycardia was the ability of atropine to reduce inhibition of the parasympathetic nerve on the heart and speed up the heart rate. Brown et al reported that the heart rate dropped to normal in a short time without other discomforts, but no data supported the long-term impact of tachycardia on patients.[10]

Vomiting as an accompaniment to nausea was described in all the included articles as a matter of concern, and different ways of addressing it would change its incidence.[26] In some trials, the rate of vomiting and nausea were lower than that of the control group, which was due to the property of antiemetics.[11-13,27] Of course, there were some examples presenting the opposite result, but the satisfaction of the patient's parents with sedation was not affected.[8,9] Finally, in our analysis, the incidence of these opposite results was not statistically significant. Furthermore, Green et al suggest age as an independent factor that might influence vomiting, noting that it was more likely to occur in older children than in those under the age of five.[28] In general, atropine use might be not be necessary to address vomiting and nausea. However, more data are needed to prove this.

For the clinical index of laryngospasm and oxygen desaturation, four studies reported outcomes and presented interpretations of their findings. [9,11,12,27] Soon after the clinical use of ketamine, researchers suggested a combination of cholinolytic drugs, whose use was associated with laryngospasm and hypersalivation.[29,30] However, the incidence of laryngospasm is related to all sedation drugs, not just ketamine, and the odds of this complication were low.[8] In addition, reports on desaturation during sedation were few, and the reason was typically inappropriate airway alignment. At the same time, airway complications were not necessarily associated with increased secretions.[12] In conclusion, reducing atropine meant

reducing potential side effects, saving on costs and management of the drug and lessening medical errors.

This meta-analysis had the following strengths. First, this was the first meta-analysis to study the use of atropine in pediatric sedation with ketamine and to discuss the safety of atropine. This work seeks to provide more reliable evidence for clinical practice. Second, we adopted strict criteria for the screening of articles so that the data reported were reasonable and reliable. Limitations included the number of included patients, the unclear risk of bias of the RCTs, and the lack of a subgroup analysis to explore whether age influences the effect of sedation.

Conclusions

With this analysis of the current evidence, ketamine plus atropine may reduce hypersalivation. No differences were observed between the two groups for nausea, vomiting, desaturation, agitation and laryngospasm. However, injection of atropine improved the appearance of rash and tachycardia when compared with using ketamine only. The current quality and quantity of experiments for this study are still relatively small. More large-scale, multicenter, high-quality studies are needed to confirm this result.

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Tables –

Table 1. Quality assessment score of the retrospective study

Table 2. Summary of the characteristics of the included studies

Figure legends –

Fig 1. Flowchart of the study selection process

Fig 2a. Risk of bias assessment of each included study-Risk of bias graph

Fig 2b. Risk of bias assessment of each included study-Risk of bias summary

Fig 3. Forest plot diagram showing the incidence of nausea

Fig 4. Forest plot diagram showing the incidence of vomiting

Fig 5. Forest plot diagram showing the incidence of desaturation

Fig 6. Forest plot diagram showing the incidence of hypersalivation

Fig 7. Forest plot diagram showing the incidence of rash

Fig 8. Forest plot diagram showing the incidence of tachycardia

Fig 9. Forest plot diagram showing the incidence of agitation

Fig 10. Forest plot diagram showing the incidence of laryngospasm

Table1. Quality assessment score of the retrospective study

Quality assessment for retrospective trial	Chong, J. H 2013
A clearly stated aim	2
Inclusion of consecutive patients	0
Prospective data collection	2
Endpoints appropriate to the aim of the study	2
Unbiased assessment of the study endpoint	2
A follow-up period appropriate to the aims of study	1
Less than 5% loss to follow up	2
Prospective calculation of the sample size	1
An adequate control group	1
Contemporary groups	0
Baseline equivalence of groups	2
Adequate statistical analyses	2

Table 2: Summary of characteristics in the studies included

Author/year	Asadi, P 2013	Lee, J. S 2012	Kye, Y. C 2012	Chong, J. H 2013	P Heinz 2006
Country	Iran	Korea	Korea	Singapore	UK
Study design	RCT	RCT	RCT	Retro	RCT
No.patients	100/100	138/110	68/72	164/119	44/39
(Atropine/ Placebo)					
Age (mean, years)	7.0/7.1	2.0 /2.1	2.9/2.6	5.3/ 5.5	3.3/3.8
(Atropine/ Placebo)					
Male(%)	65/72	73.7/64.5	70.6/68.1	61.5/ 69.7	63.6/74.4
Weight (mean, Kg)	23.2/23.0	12.8/13.1	Nr	Nr	15.7/14.9
(Atropine/ Placebo)					
ASA scores	I or II	I or II	I	I	Nr
	1 mg/kg K plus 0.01	4 mg/kg K plus	2 mg/kg K plus	3 to 4 mg/kg K	4 mg/kg K plus
	mg/kg A (IV)	0.01 mg/kg A	0.01 mg/kg A	plus 0.02 mg/kg A	0.01 mg/kg A
Interventions		(IM)	(IV)	(IM)	(IM)
	1 mg/kg K plus distilled	4 mg/kg K alone	2 mg/kg K plus	3 to 4 mg/kg K	4 mg/kg K plus
	water (IV)	(IM)	normal saline	(IM)	Normal saline
			(IV)		(IM)

K: ketamine, A: atropine, RCT: randomized controlled trial, Retro: retrospective study, ASA scores: American

Society of Anesthesiologists scores, IM: intramuscularly, IV: intravenously, Nr: Not reported.

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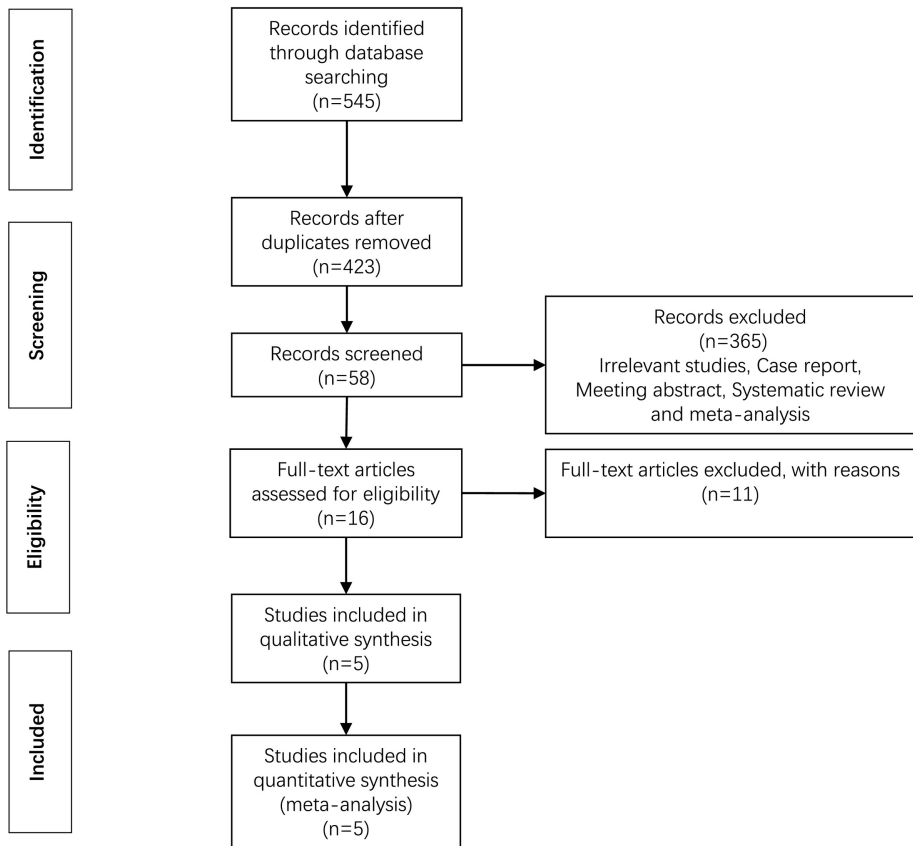


Figure 1

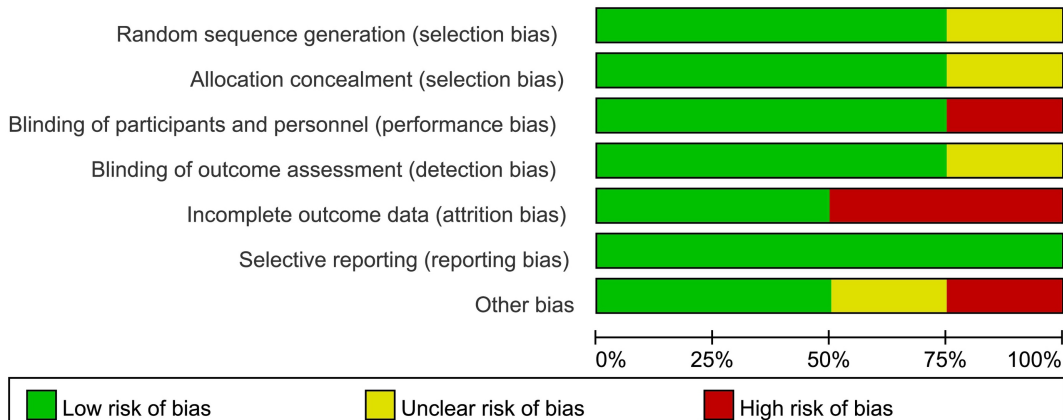


Figure 2a

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
Asadi P 2013	+	+	+	+	+	+	+
Kye YC 2012	+	?	-	?	-	+	+
Lee JS 2012	+	+	+	+	-	+	?
P Heinz 2006	?	+	+	+	+	+	-

Figure 2b

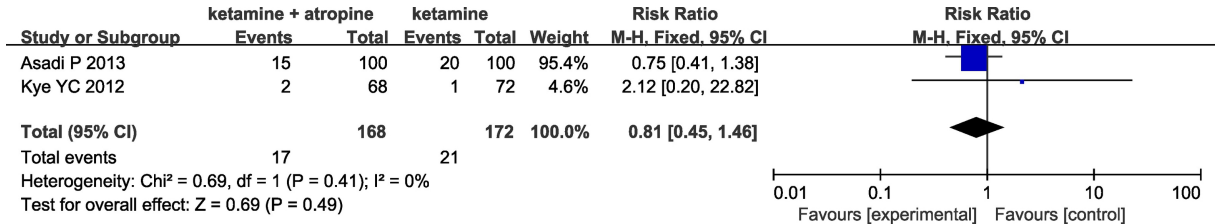


Figure 3

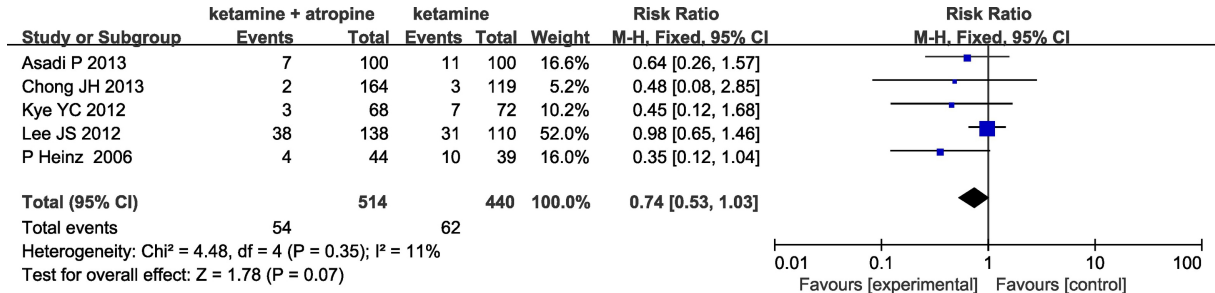


Figure 4

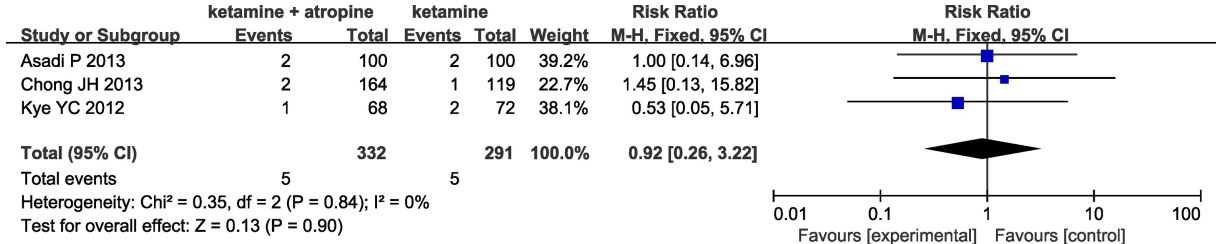


Figure 5

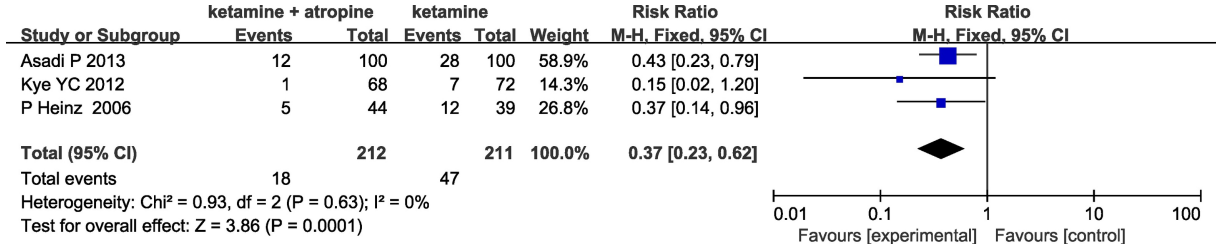


Figure 6

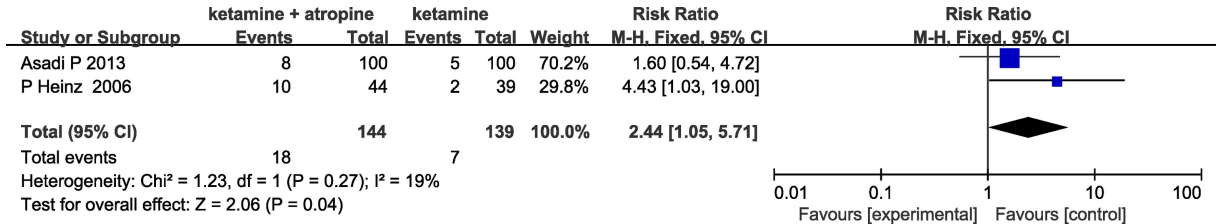


Figure 7

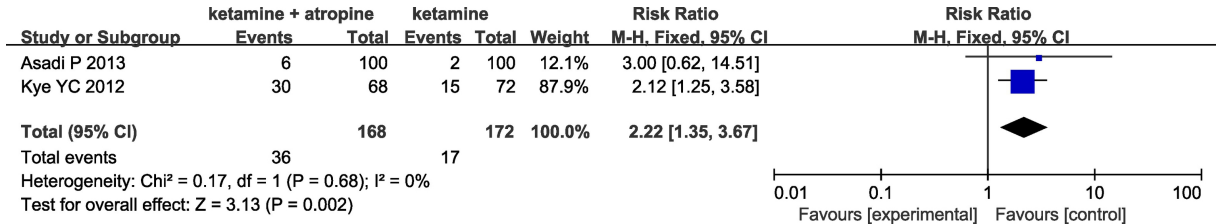


Figure 8



Figure 9

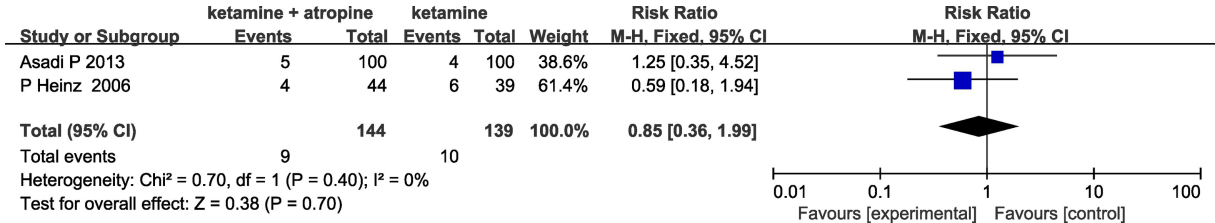


Figure 10